

## Exemplar: Enterprise Data Architecture and Operations

This document presents an exemplar solution to the lab, 'Enterprise Data Architecture and Operations'. Please remember that this is just one point of view and not a definitive solution.

### Solution

#### Task 1: Understand the requirements

Step 1: Clearly outline the system's primary features and capabilities

| Primary features of the system |
|--------------------------------|
| Account opening and closure    |
| Deposit and withdrawal         |
| FD/RD management               |
| Account type differentiation   |

Step 2: Specify the data requirements necessary for the system to execute each operation effectively.

| Necessary data points  |
|------------------------|
| Account holder details |
| Transaction history    |
| Balance details        |

#### Task 2: Identify the key identities

Clearly determine and define the primary components or building blocks of the system.

| Key entities | Description                                  | Key attributes                                 |
|--------------|----------------------------------------------|------------------------------------------------|
| Customer     | Stores customer details.                     | Name, contact, address, ID proof               |
| Account      | Tracks account details.                      | Account number, type, balance, status          |
| Transaction  | Logs deposits, withdrawals, and transfers.   | Transaction ID, date, amount, type             |
| FD_RD        | Manages fixed and recurring deposit details. | Deposit ID, term, interest rate, maturity date |
| Branch       | Represents the bank's branches.              | Branch ID, location, contact                   |

|              |                                                                                                   |                                           |
|--------------|---------------------------------------------------------------------------------------------------|-------------------------------------------|
| Employee     | Tracks employees managing operations.                                                             | Employee ID, role, branch                 |
| Account_Type | Differentiates between account types (e.g., savings, checking, fixed deposit, recurring deposit). | Account types (savings, checking, FD, RD) |

### Task 3: Define attributes for each entity

Identify and specify the key characteristics or properties that describe each entity.

| Key entities | Attributes and data types                                                                                                                                                                                                                                                             |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Customer     | <ul style="list-style-type: none"> <li>- Customer_ID (Primary Key)</li> <li>- Name (VARCHAR)</li> <li>- Address (VARCHAR)</li> <li>- Contact_Info (VARCHAR)</li> <li>- ID_Proof (VARCHAR)</li> </ul>                                                                                  |
| Account      | <ul style="list-style-type: none"> <li>- Account_ID (Primary Key)</li> <li>- Customer_ID (Foreign Key)</li> <li>- Branch_ID (Foreign Key)</li> <li>- Account_Type_ID (Foreign Key)</li> <li>- Balance (Decimal)</li> <li>- Opening_Date (Date)</li> <li>- Status (VARCHAR)</li> </ul> |
| Transaction  | <ul style="list-style-type: none"> <li>- Transaction_ID (Primary Key)</li> <li>- Account_ID (Foreign Key)</li> <li>- Date (Date)</li> <li>- Amount (Decimal)</li> <li>- Transaction_Type (VARCHAR)</li> <li>- Description (VARCHAR)</li> </ul>                                        |
| FD_RD        | <ul style="list-style-type: none"> <li>- FD_RD_ID (Primary Key)</li> <li>- Account_ID (Foreign Key)</li> <li>- Type (VARCHAR)</li> <li>- Term (Number)</li> <li>- Interest_Rate (Decimal)</li> <li>- Start_Date (Date)</li> <li>- Maturity_Date (Date)</li> </ul>                     |
| Branch       | <ul style="list-style-type: none"> <li>- Branch_ID (Primary Key)</li> <li>- Name (VARCHAR)</li> <li>- Location (VARCHAR)</li> <li>- Contact (VARCHAR)</li> </ul>                                                                                                                      |
| Employee     | <ul style="list-style-type: none"> <li>- Employee_ID (Primary Key)</li> <li>- Branch_ID (Foreign Key)</li> <li>- Name (VARCHAR)</li> </ul>                                                                                                                                            |

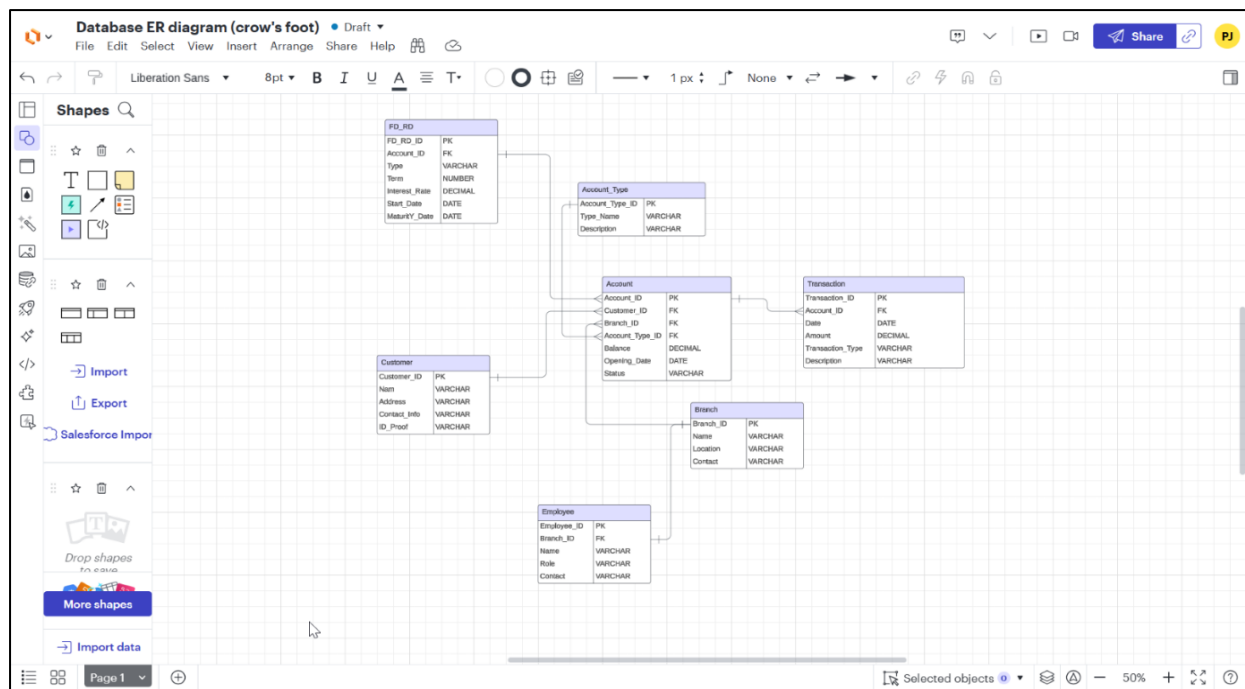
|              |                                                                                                                                                                                    |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | <ul style="list-style-type: none"> <li>- Role (VARCHAR)</li> <li>- Contact (VARCHAR)</li> </ul>                                                                                    |
| Account_Type | <ul style="list-style-type: none"> <li>- Account_Type_ID (Primary Key)</li> <li>- Type_Name (e.g., Savings, Checking, FD, RD) (VARCHAR)</li> <li>- Description(VARCHAR)</li> </ul> |

#### Task 4: Establish the relationship between entities

Clearly define how the entities within the system are connected or interact with one another.

| Entities | Relationships      | Cardinality |
|----------|--------------------|-------------|
| Customer | can have multiple  | 1:N         |
| Account  | can have multiple  | 1:N         |
| Account  | may be linked to   | 1:N         |
| Account  | is associated with | N:1         |
| Branch   | can have multiple  | 1:N         |
| Employee | belongs to one     | 1:1         |

#### Task 5: Create Entity Relationship Diagram using Lucidchart



**Note:** By applying data modeling principles, designing a normalized data model, and creating an Entity Relationship Diagram (ERD) in Lucidchart to visually represent the entities, attributes, and relationships within a processing system, you can enhance your understanding of data organization and improve the effectiveness of system design and analysis.